

RESEARCH INTERESTS

Bio-Inspired Robotics; Compliant Actuation; Human-like Locomotion; Reinforcement Learning; Motion Planning & Control

EDUCATION

Zhejiang University (ZJU)

Aug. 2024 – Jun. 2028 (expected)

ZJU–UIUC Institute | B.Eng. in Mechanical Engineering

- **GPA:** 3.84/4.00; Top 10%.
- **Honors & Awards:** Third-Class University Scholarship (Oct. 2025, Jul. 2026); Academic Excellence Award (Sep. 2025)

University of Illinois Urbana–Champaign (UIUC)

Aug. 2024 – May. 2028 (expected)

The Grainger College of Engineering | B.S. in Mechanical Engineering

- **GPA:** 4.00/4.00.
- **Honors & Awards:** Dean's List (2024–2026, 4 academic semesters).
- **English Proficiency:** TOEFL 5.5/6, GRE 330/340.

RESEARCH EXPERIENCE

Elastic Subsystem for Compliant Bio-Inspired Manipulators

RA @ Intelligent Robot Vision & Control Lab | Advisor: Liangjing YANG (ZJUI) | Oct. 2025—Present

- Developing a human upper-limb-inspired spring–cable elastic subsystem for robotic elbow actuation, enabling mechanical-energy recovery during repetitive motion and passive mitigation of transient impact loads.
- Designing and prototyping the elbow-integrated spring–cable elastic system (SCES), while optimizing spring placement, cable routing, mechanism geometry, and interfaces through CAD-based design and kinematic modeling.
- Validating the mechanism through cyclic elbow-motion and impact experiments, quantifying energy consumption, joint tracking performance, elastic force transmission, and peak impact torque.

Bio-Inspired Elastic Bipedal Robotics: Mechanism Design and Physics-Aware Reinforcement Learning

RA @ State Key Lab of Fluid Power & Mechatronics | Advisors: Tao LIU (ZJU), Jingang YI (Rutgers) | May. 2025—Present

- Co-designing the STERS mechanical structure, an active-toe-coupled spring–tendon mechanism for cross-joint ankle–toe energy storage, transfer, and release during heel-to-toe walking.
- Modeling tendon-induced ankle–toe torques, optimizing spring stiffness, and developing a physics-aware RL controller in Isaac Lab with a human-like contact-stage gait that actively exploits STERS for energy recirculation.
- Validating STERS on BioLeg and Roboto Origin; hardware ablations have reduced leg energy consumption by 44.3% and 36.3% and peak ankle torque by approximately 75%, while improving impact recovery and heading preservation.

PUBLICATIONS

- **Y. Zhang**, et al. “*STERS: An Active-Toe-Coupled Spring-Tendon Energy Recirculation System for Efficient Bipedal Locomotion*” (*IEEE/ASME Transactions on Mechatronics*, Under review)
- **Y. Zhang**, et al. “*BioArm: A Biomimetic Robotic Arm with a Human Upper-Limb-Inspired Spring–Cable Elastic System*” (*IEEE International Conference on Robotics and Automation System*, Under review)

INDUSTRY EXPERIENCE

LimX Dynamics — Robotics Engineer (Intern), Hangzhou

Jun. 2025 – Sep. 2025

- Optimized the leg structure of the LimX Oli bipedal robot through iterative mechanical design to improve adaptability on rugged terrain while meeting locomotion and control requirements.
- Developed and evaluated RL locomotion controllers on the LimX Oli platform, linking policy-training results with structural-design iterations for hardware–control co-design.

SELECTED ENGINEERING PROJECTS

- **Autonomous Line-Following and Obstacle-Avoidance Car** (ECE 110 @ ZJUI) — Integrated sensing, actuation, and control logic for autonomous line tracking and obstacle avoidance.
- **Automated Fruit-Cutting Machine** (ME 170 @ ZJUI) — Designed and prototyped an automated mechanism integrating mechanical design, actuation, and system control.
- **Remote-Controlled Automatic Bait-Dispensing Boat** (ME 270 @ ZJUI) — Developed a remote-controlled boat with an automated bait-dispensing mechanism and integrated mechanical and control subsystems.

TECHNICAL SKILLS

Skills	Python, C, C++, MATLAB; Fusion360, SolidWorks; MuJoCo, IsaacLab, IsaacSim, ROS
Languages	Chinese (native), English (proficient)